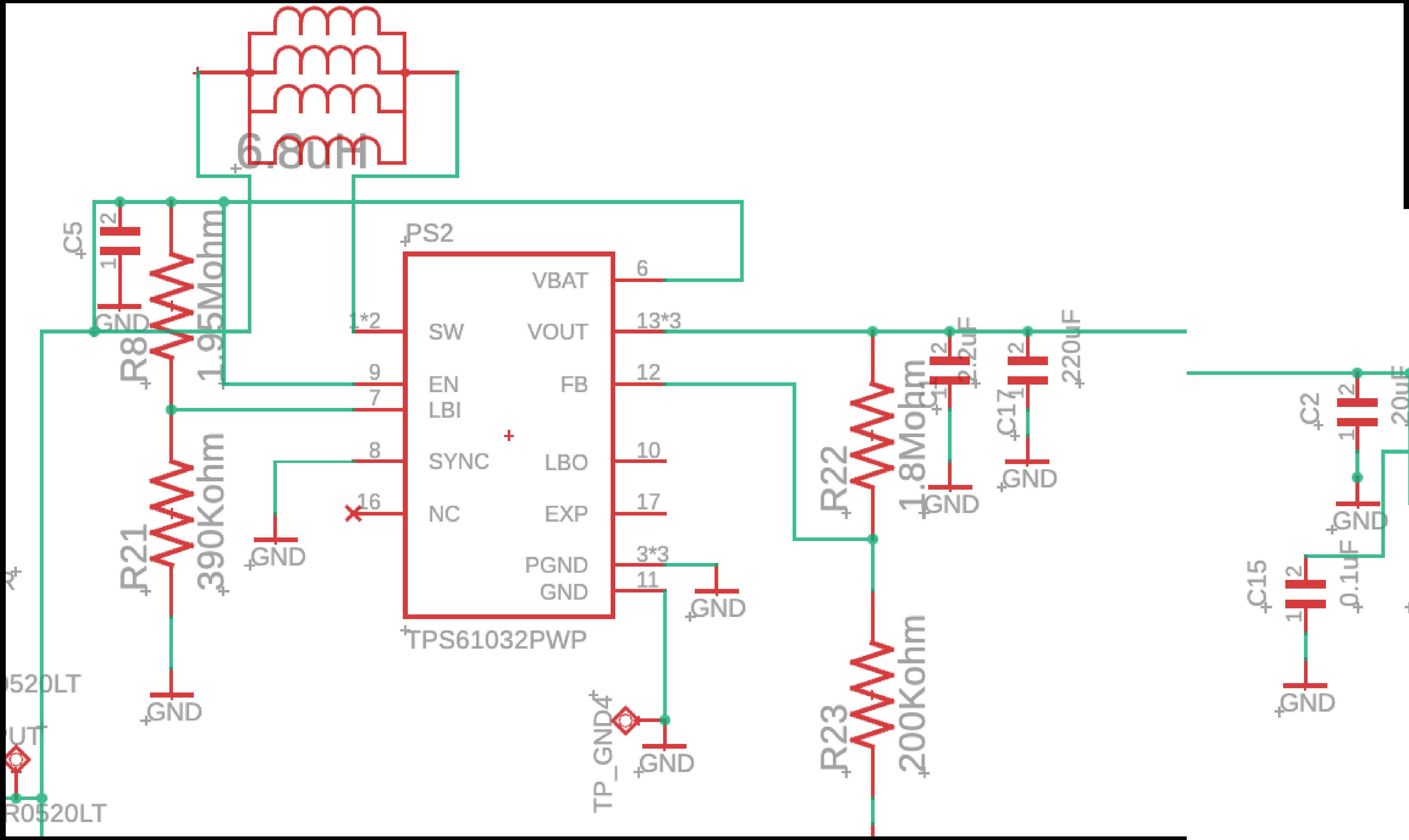
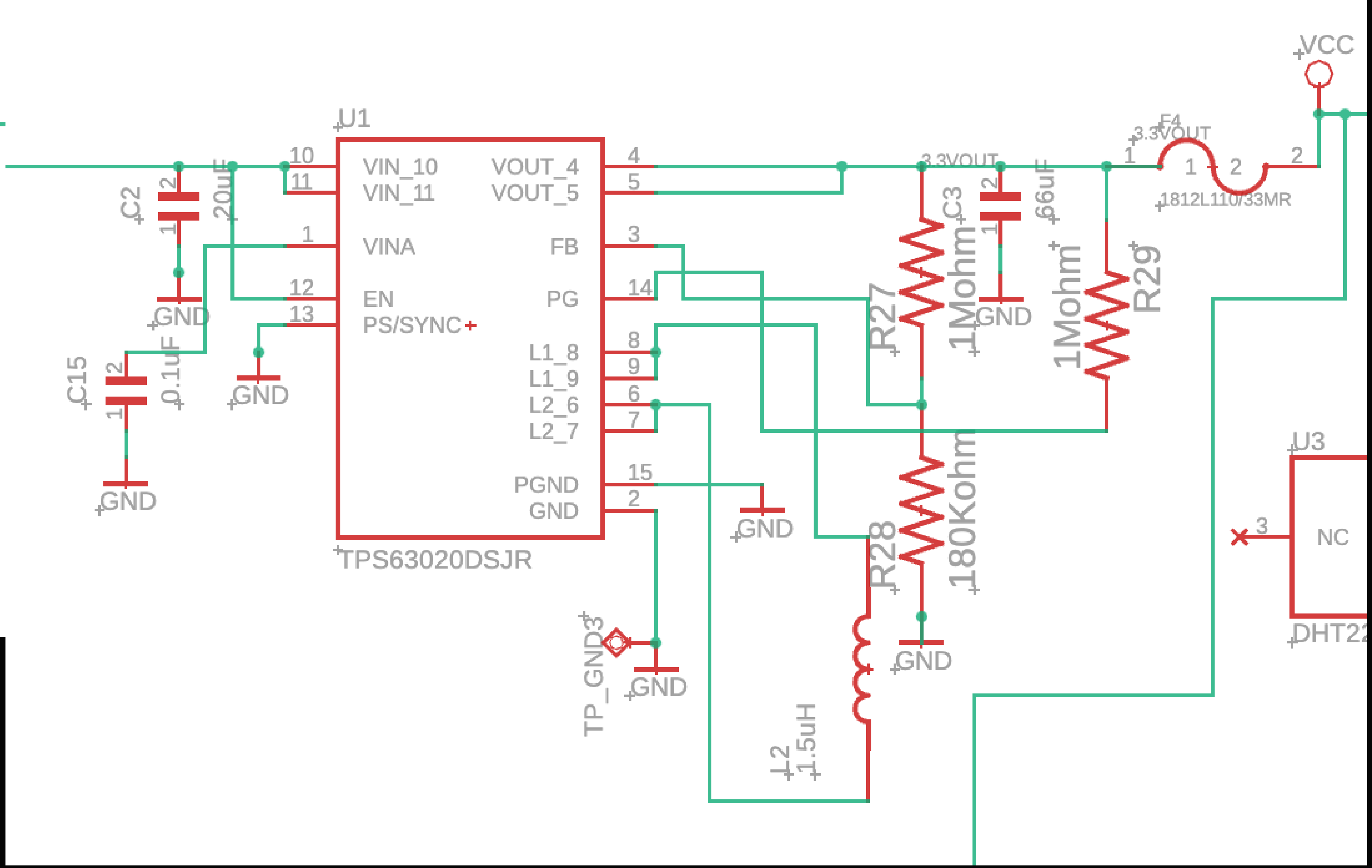
CLEAN ROOM MONITORING SYSTEM

SECOND REVISION

POWER



Booster, output 5V



Buck-Boost, output 3.3 V

Main parts in system.

Power goes to sensors + components

Both accepts voltage range of: 1.8V - 5.5V.

Output => determined by the feedback resistors

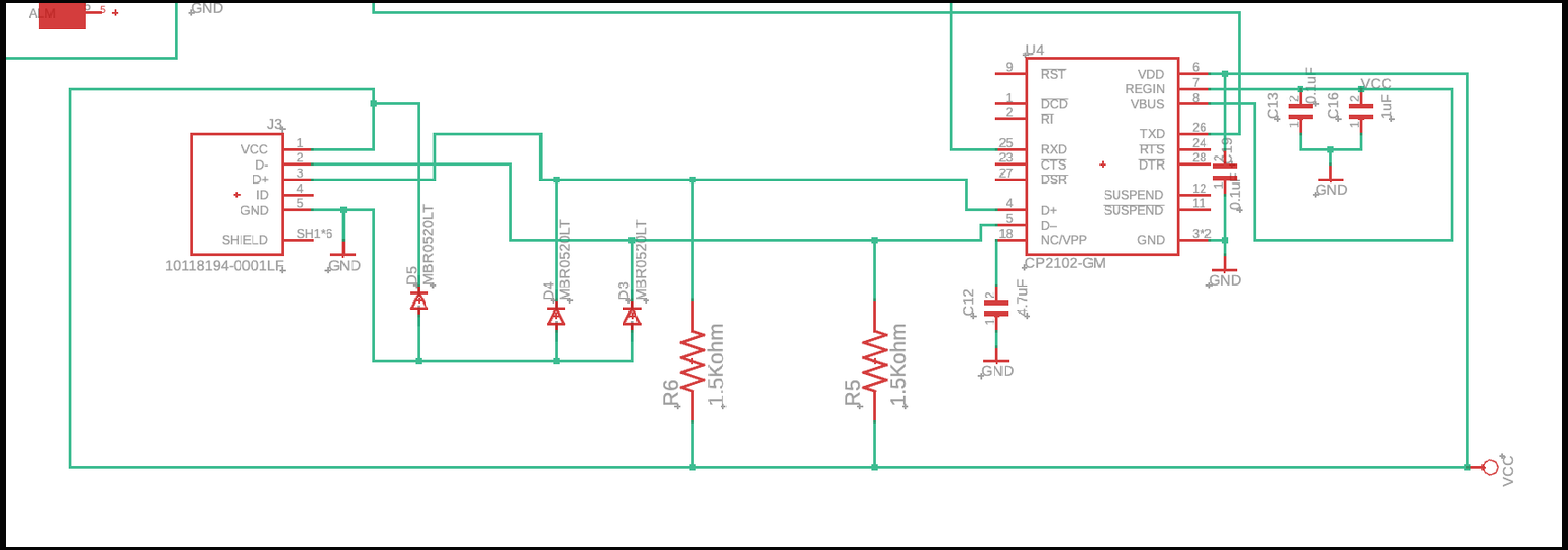
R3 = R4(Vout/1 - Vfb), Vfb = 0.5V

USB CONNECTION

CANNOT UPLOAD CODE → PORT NOT SHOWING

POSSIBLE ISSUES:

- CABLE → X
- DRIVERS INSTALLATION → X
- ADDITIONAL LIBRARIES → X



Previous schematic

retrieved from:
https://www.silabs.com/documents/public/data-sheets/CP2102-9.pdf

CP2102 GM

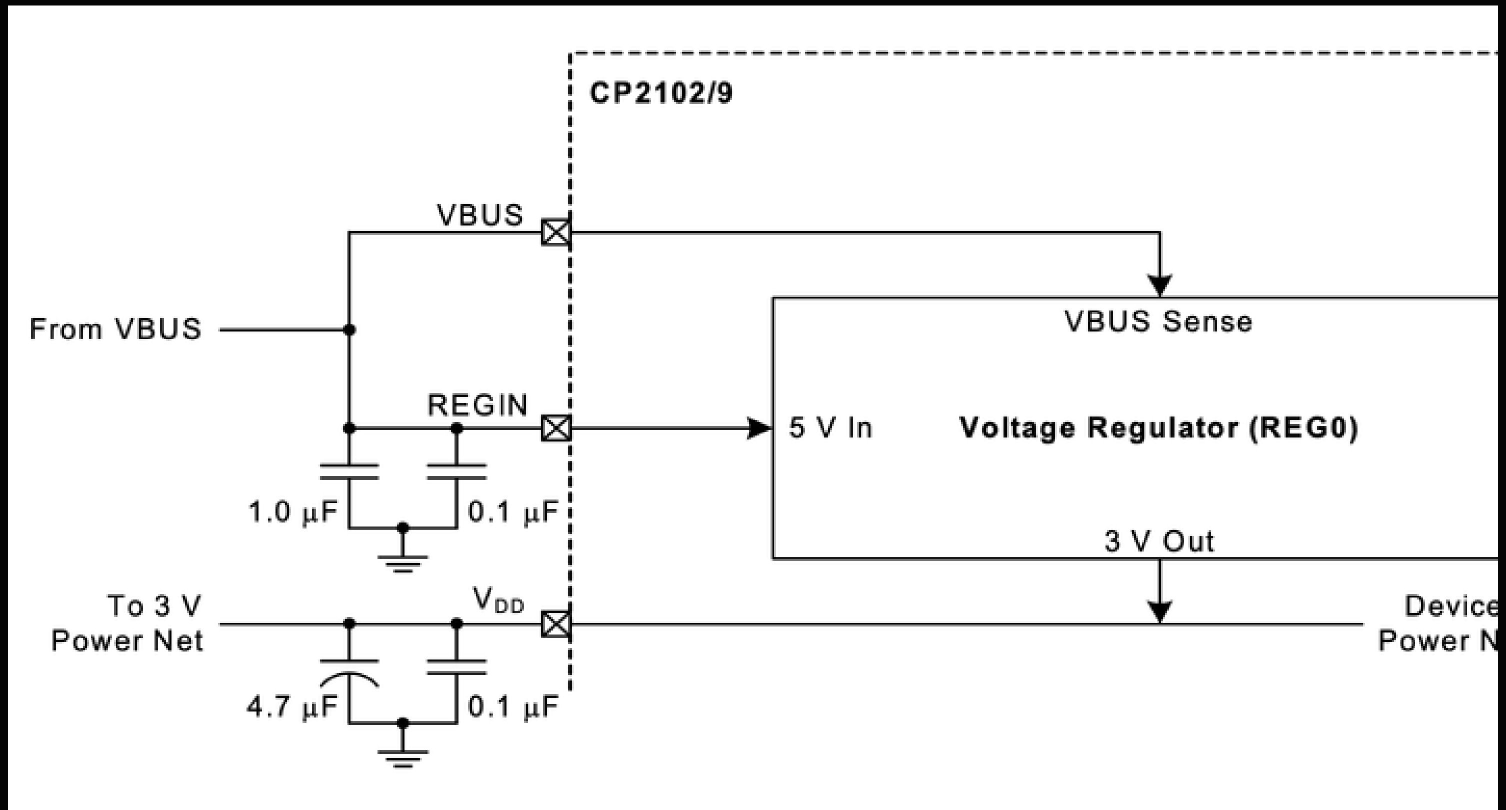
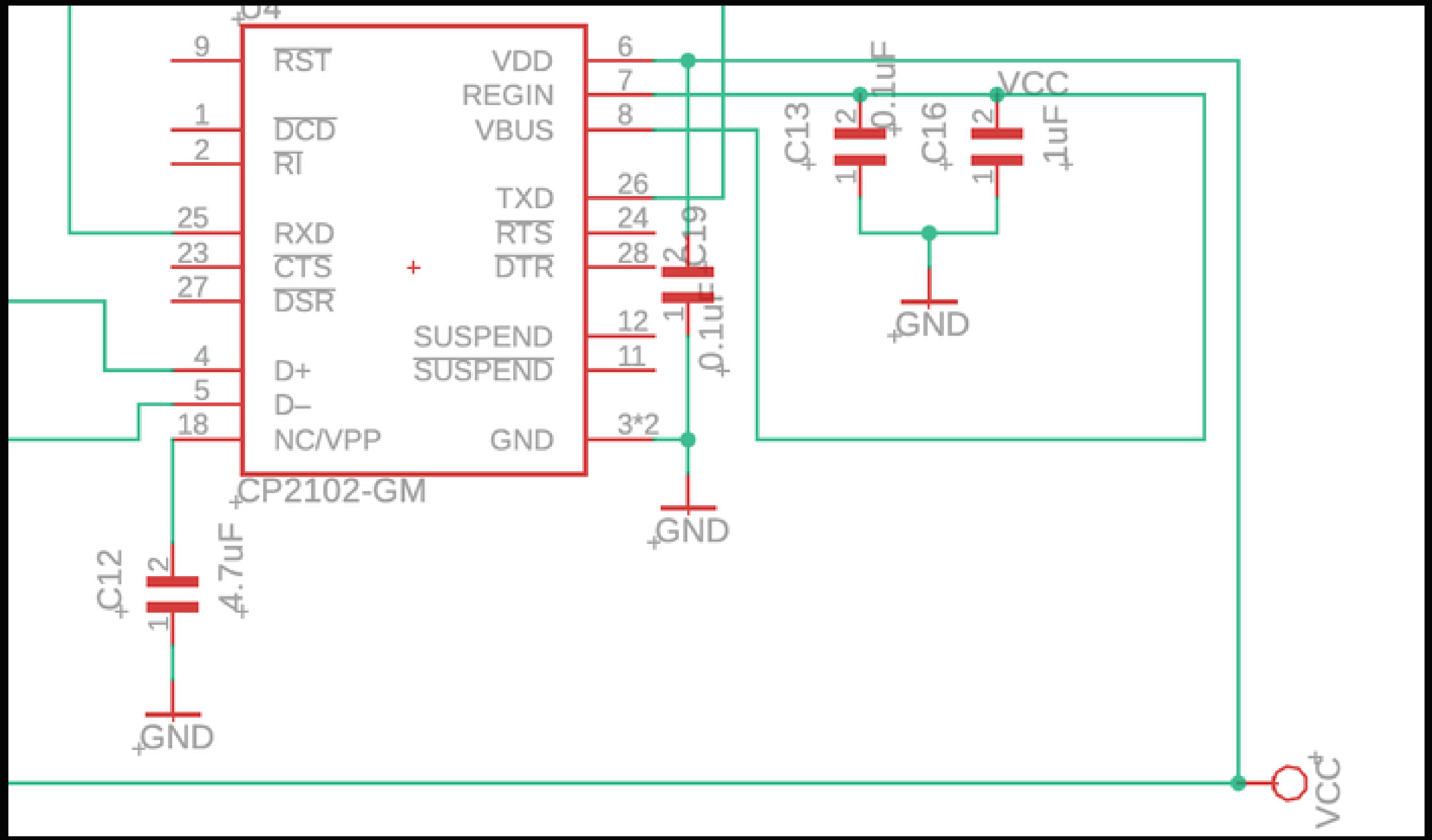


Figure 6. Configuration 1: USB Bus-Powered



Data sheet

Previous schematic

Problem:
USB is only for uploading code, not supplying power

retrieved from: https://www.silabs.com/documents/public/data-sheets/CP2102-9.pdf

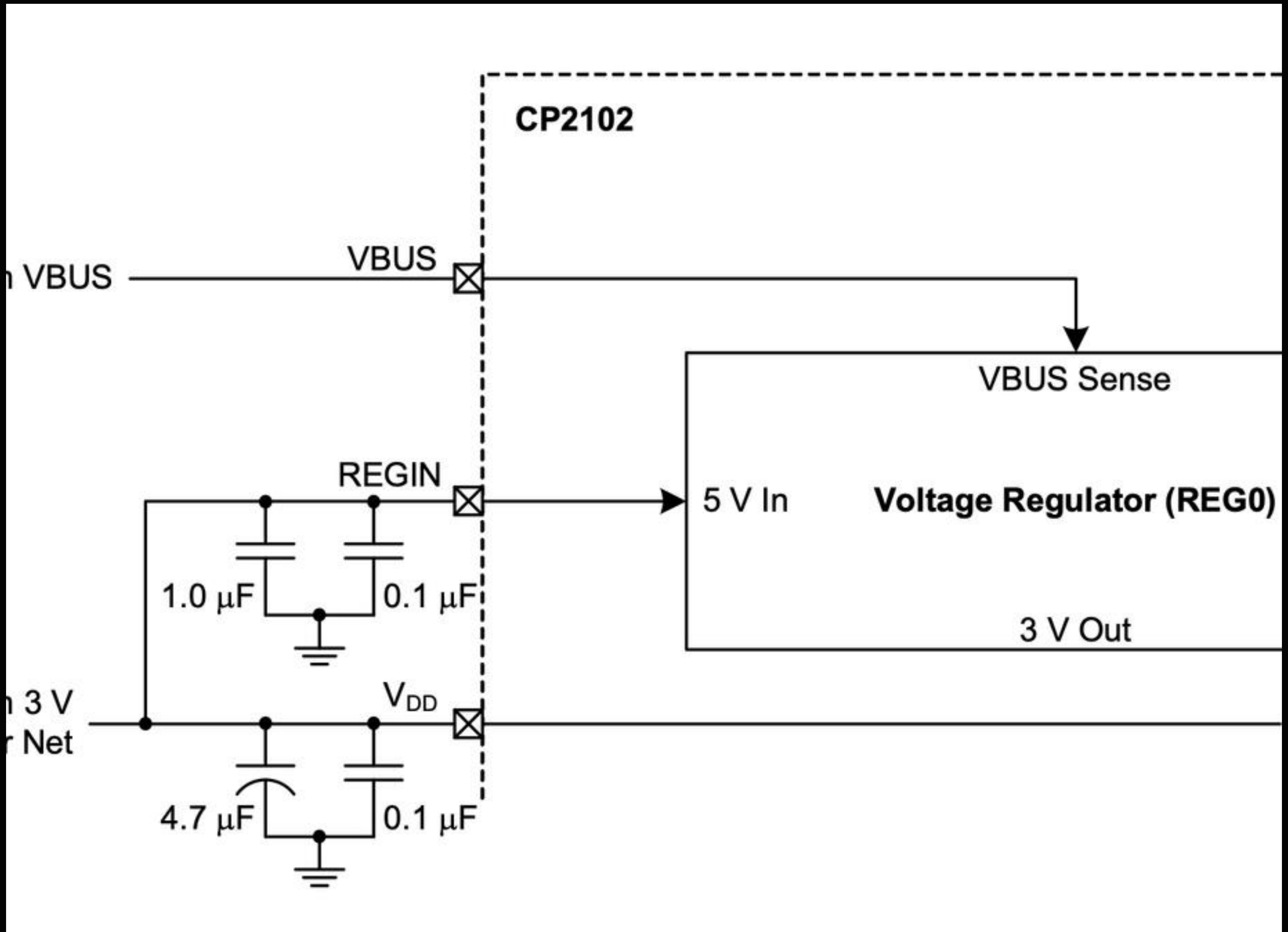
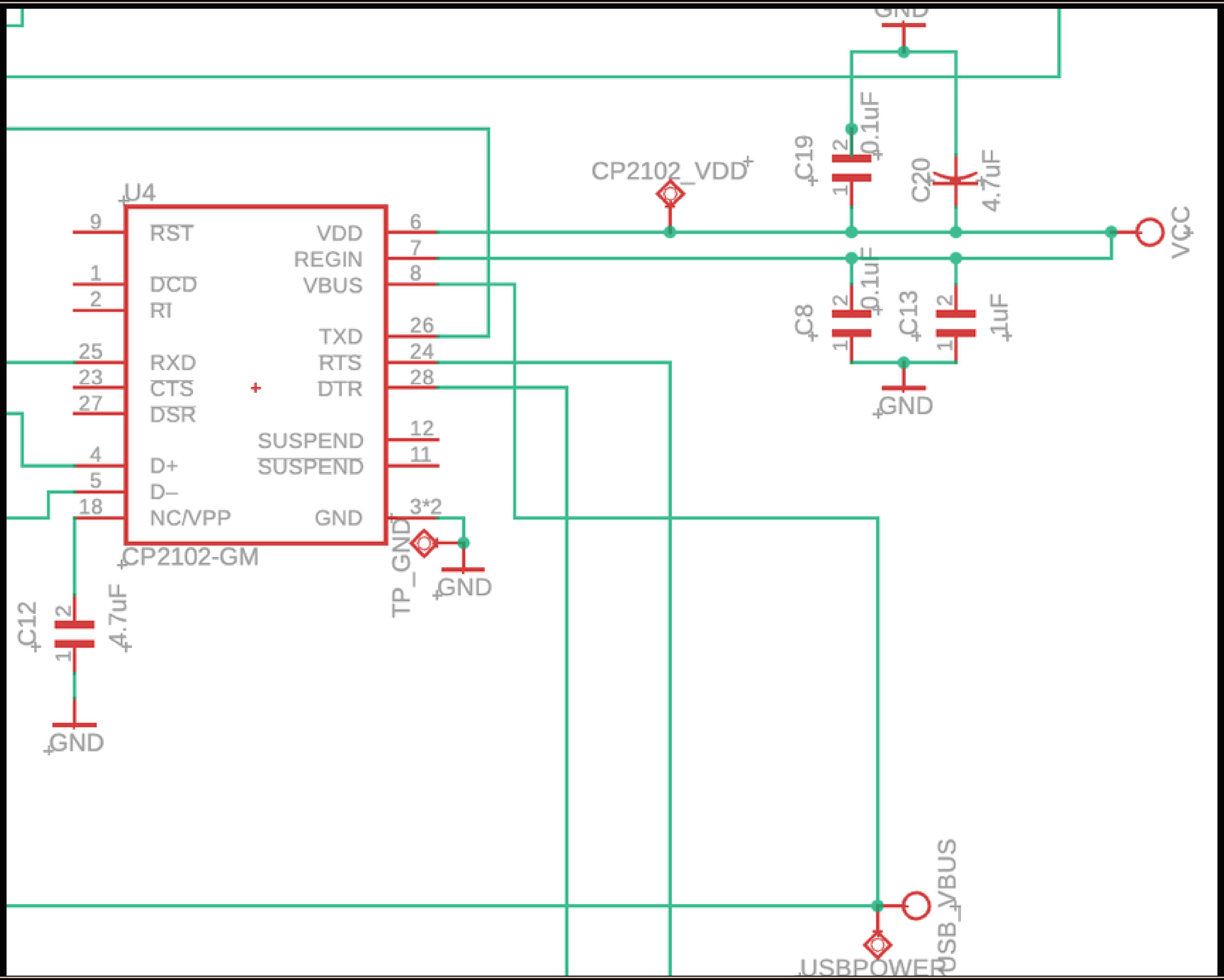


Figure 9. CP2102 Configuration 3: USB Self-Powered, Regulator Bypassed

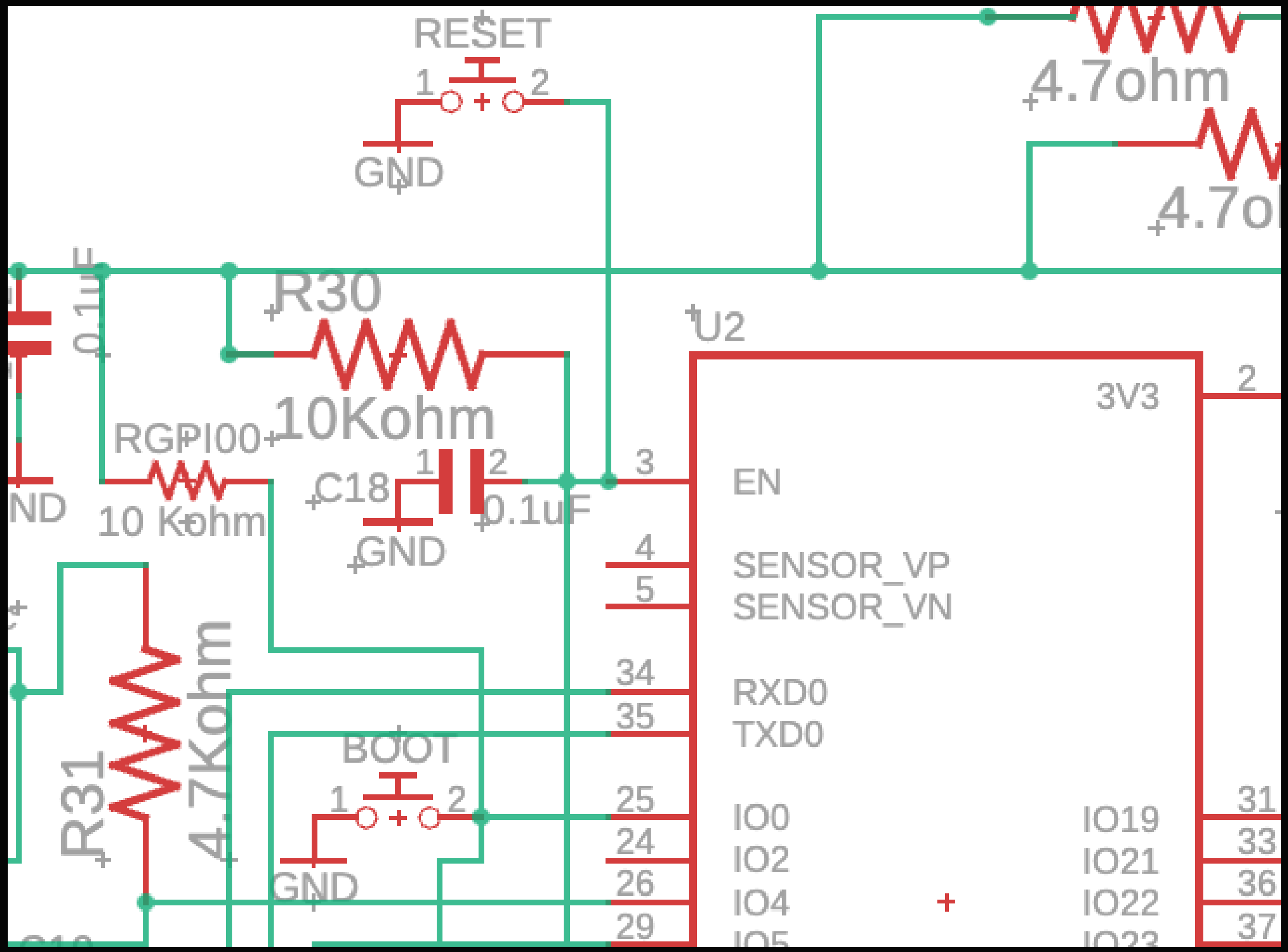


Data sheet

Current schematic

Fix:
A different USB_VBUS line,
self-powered system.

BOOT + RESET



Code Upload Problem?

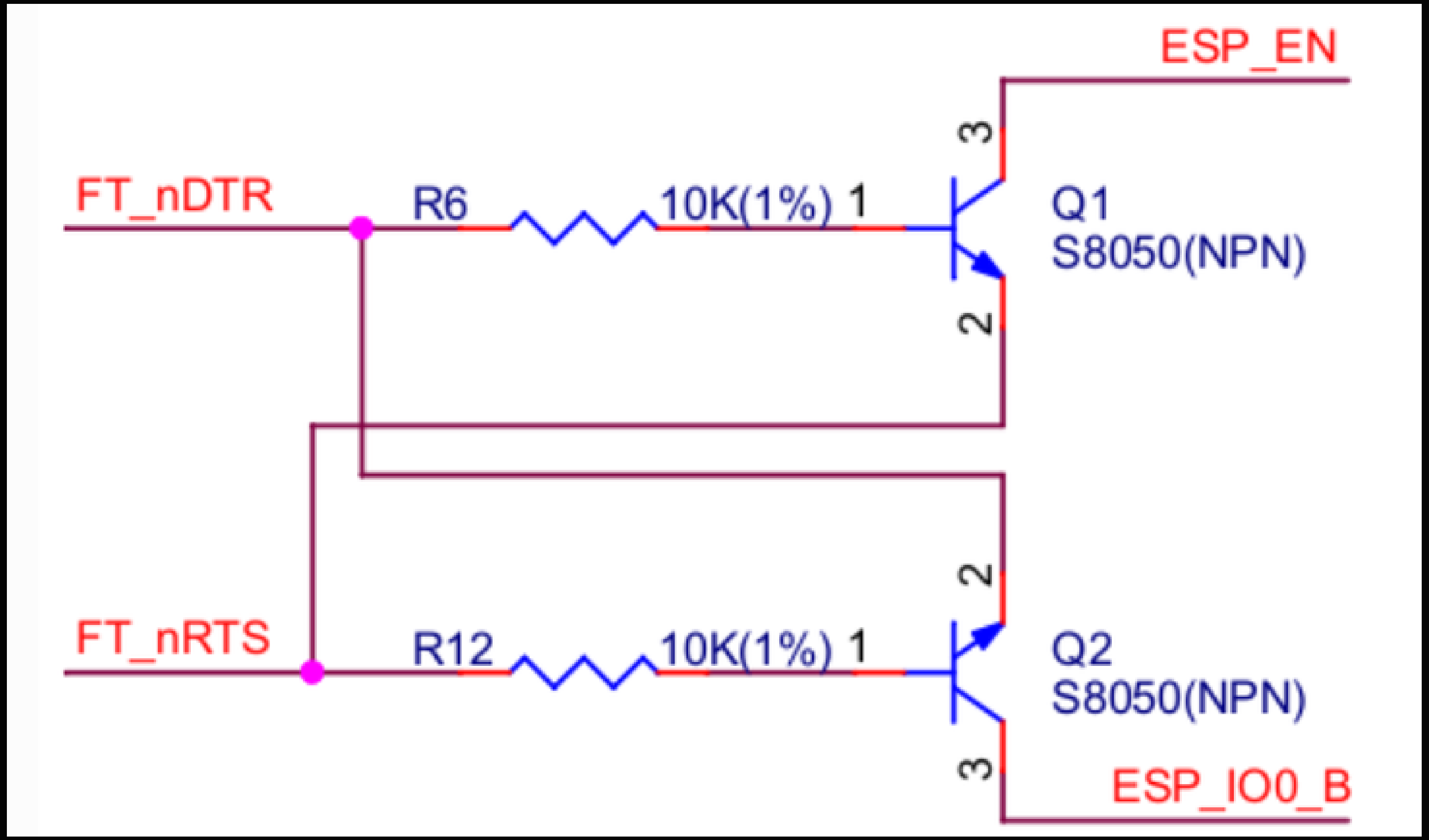
Manually force ESP32 to enter bootloader mode to successfully upload code

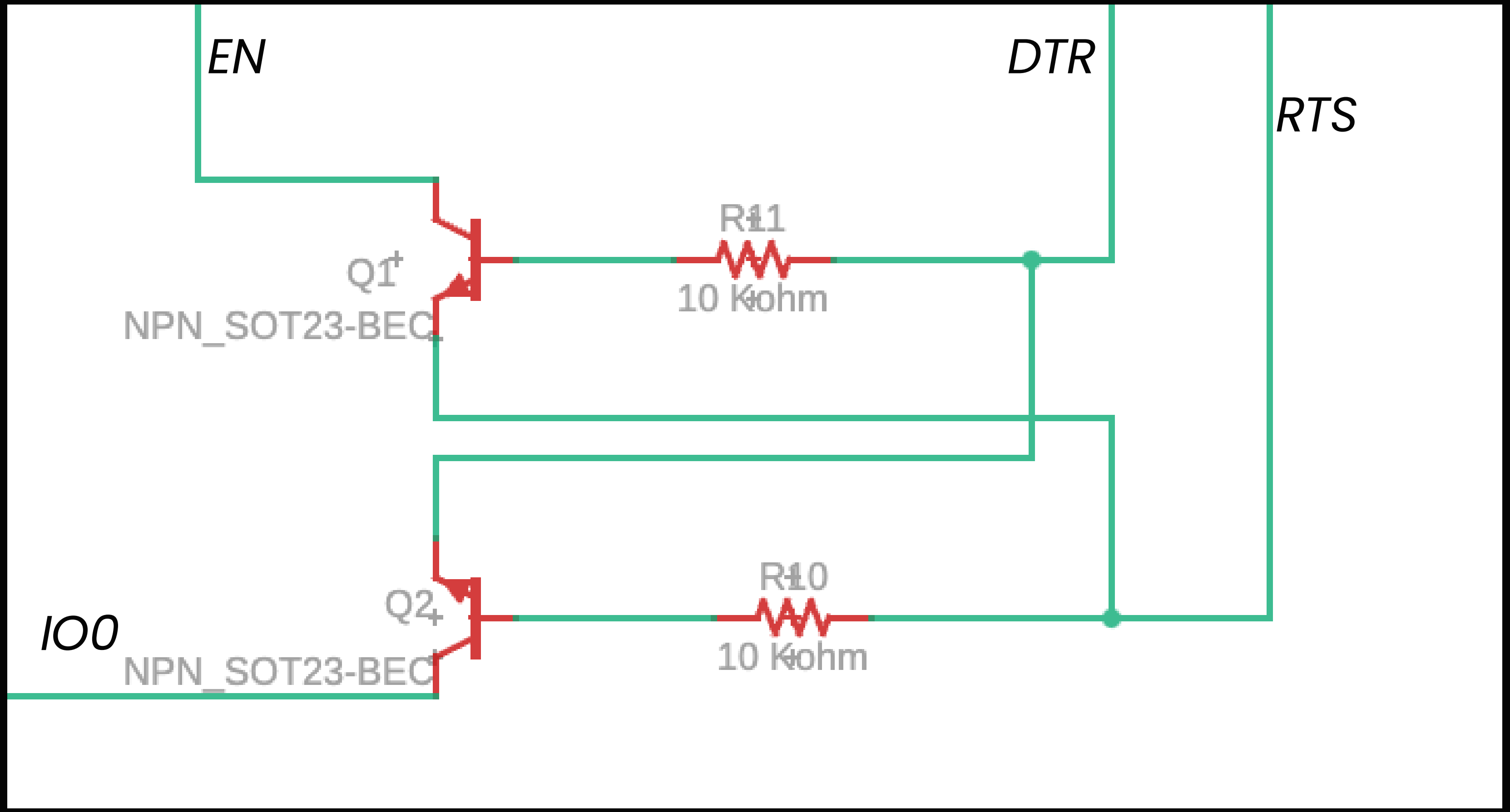
How?

1. Click Upload on Arduino
2. Wait until you see:

Connecting...

1. Perform the sequence:
   - Hold BOOT
   - Press and release RESET
   - Release BOOT when you see “writing...”





AUTO PROGRAMMING

For all ESP modules, the reference image is recommended to enter boot loader mode.

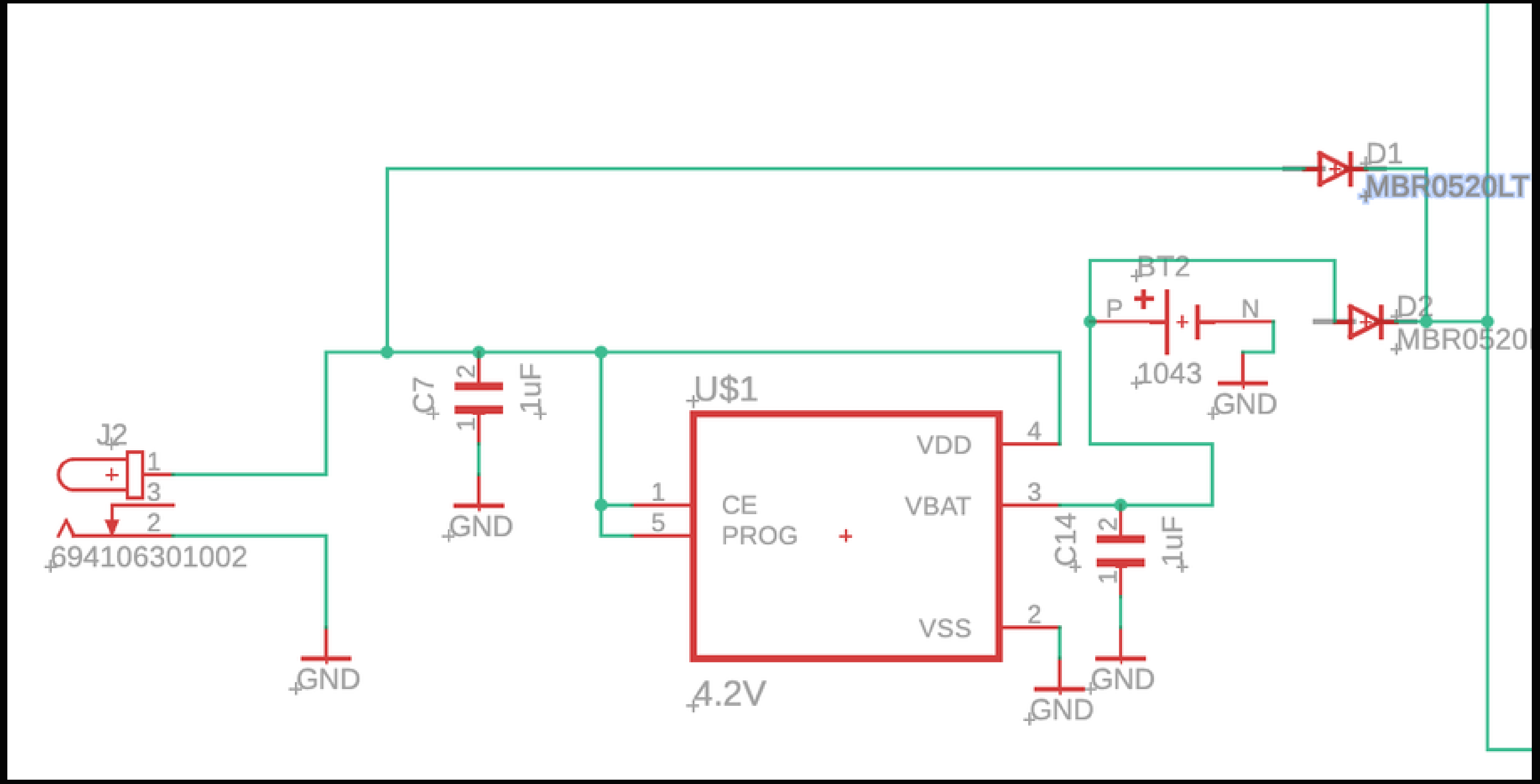
Eliminates the need to manually press the BOOT and RESET buttons every time you upload code.

retrieved from: https://docs.espressif.com/projects/esp-iot-solution/en/latest/hw-reference/ESP-Prog_guide.html

POWER SUPPLY

Schotky diodes determine the voltage through the intersection.

Higher voltage will pass.



Power Supply Architecture

DC JACK

Direct ~5V supplied to system

Initial stage and fully charged stage: still supplies voltage

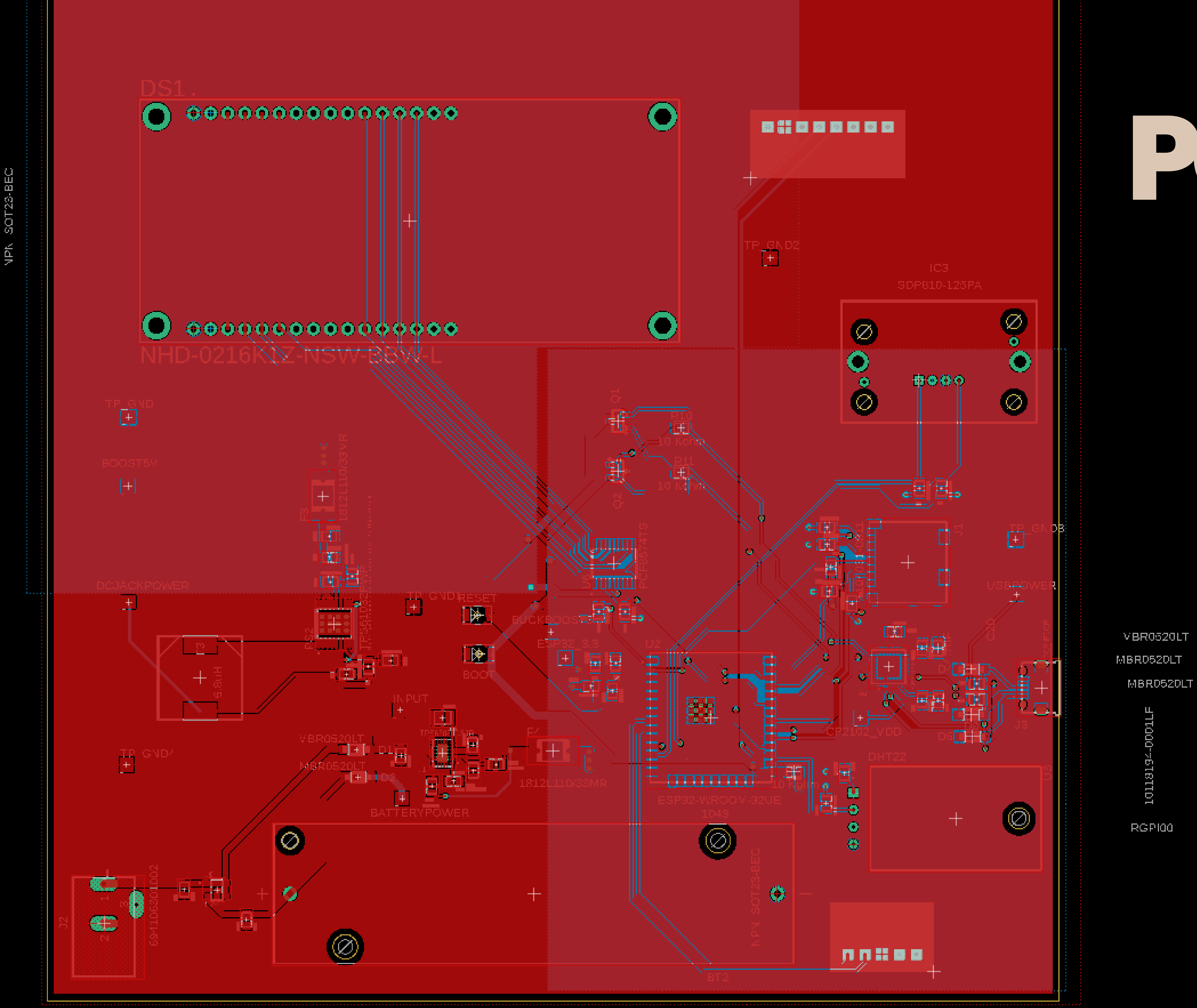
Battery stops discharging at 3.0V to protect cell

BATTERY

Charges while DC is connected.

Supplies power once DC is disconnected, reliable in power outages.

3.0V - 4.2V discharge



PCB BOARD

TEST PADS

Reset and Boot Buttons

Auto Programming

Alternative Wirings

CODE + BLYNK

```
//include the SDP810 - differential pressure -> done

//ISO Class 7 Specs

//System State Variables

float temp = 0;
float hum = 0;
float oxygen = 0;
int pm25 = 0;
int pm10 = 0;
float pressure = 10;
//temporary until better logic to update it

bool RoomisClean = true;

//Temperature
const float MIN_TEMP = 18.0;
const float MAX_TEMP = 25.0;
```

Extract of Arduino code

- CLEANED
- DEBUGGED
- IMPROVED

- REMOTE NOTIFICATION ENABLED WITH BLYNK.IO

NEXT PAGE

CleanRoom_Monitor	UEE4XqkVBlCT9Dm1RtXsI6M1kZIp6g1q	stobd23@icloud.com (you)	Online

```
#define BLYNK_TEMPLATE_ID "TMPL6YahddNDh"
#define BLYNK_TEMPLATE_NAME "CleanRoom"
#define BLYNK_AUTH_TOKEN "UEE4XqkVBlCT9Dm1RtXsI6M1kZIp6g1q"
```

FURTHER STEPS

- CHECK HARDWARE
- UPLOAD CODE, RUN IT
- TEST SENSOR READINGS
- ENCLOSURE DESIGN FOR COMPONENTS